

		and constant.
9	A	By conservation of linear momentum, taking rightward as positive, $m_X u_X = m_X v_X + m_Y v_Y$ $(1.0)(4.0) = 1.0 v_X + (10.0)(0.50)$ $v_X = -1.0 \text{ m s}^{-1} \text{ (to the left)}$ Difference in kinetic energy of the system $= KE_i - KE_f$ $= \frac{1}{2}(1.0)(4.0)^2 - \frac{1}{2}(1.0)(-1.0)^2 - \frac{1}{2}(10.0)(0.50)^2$ $= 6.25$ $\approx 6.3 \text{ J}$
10	B	Force on satellite